

IDENTIFIABILITY-AWARE JOINT qPCR PANELS FOR COMPOSITIONAL CELL-FRACTION DECONVOLUTION

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ABSTRACT

Estimating the composition of mixed biological samples from a small qPCR panel requires recovering multiple cell-type fractions simultaneously, while accounting for compositional closure, reference-gene normalization, and confusion between biologically related populations. This is difficult because marker expression signatures can be redundant or poorly identifiable, and because discovery-derived panels must generalize to independently generated mixtures spanning rare and dominant cell abundances. We develop an identifiability-aware joint panel-design framework that constructs cluster-specific signatures using discovery cells, excludes reference genes from marker selection, and uses a forward-selection objective combining target discrimination, cross-panel redundancy control, and joint signature identifiability. Held-out pseudobulk mixtures are then quantified jointly with non-negative least squares followed by simplex normalization, with best-single-marker and one-vs-rest panel procedures retained as baselines. Across the completed experiments, correcting a global marker-allocation failure restored the intended per-cluster panel budget and improved macro MAE from 0.1778 to 0.0455, macro RMSE from 0.2428 to 0.0669, and macro R^2 from -2.3905 to 0.7694, while macro ROC AUC reached 0.9814. However, the discrimination-only, redundancy-aware, and full identifiability-aware configurations produced essentially identical reported aggregate metrics, so these experiments do not establish an ablation-sensitive advantage for the structural selection terms. The results therefore demonstrate the importance of correctly allocating markers for joint identifiability and provide a reproducible framework for evaluating coherent compositional deconvolution, while highlighting the need for independent baseline assembly and more discriminating simulations to assess the scientific hypothesis.

1 INTRODUCTION

Quantifying the composition of a mixed biological sample is important for interpreting heterogeneous tissues, validating sample preparation, and monitoring changes in cell populations. In many practical settings, however, measurements are limited to a small qPCR panel rather than a complete transcriptomic profile. The resulting task is to estimate the fractions of several cell types from a compact set of normalized gene measurements, rather than to classify one target population independently of the others. A useful solution should therefore produce estimates that are accurate for individual populations, coherent as a composition, and interpretable in terms of the genes selected for the assay.

The problem is challenging for several related reasons. Cell fractions are compositional: increasing the estimated contribution of one population necessarily affects the allocation of the remaining mass. Marker genes can also be expressed by multiple biologically related clusters, making their signatures redundant or difficult to distinguish in mixtures. Reference genes introduce an additional normalization requirement and should not be selected as discriminatory markers. Finally, a panel selected from discovery cells must be applied to independently generated holdout mixtures, so selection and estimation procedures must avoid relying on information from the evaluation samples. These constraints mean that strong one-target-versus-rest discrimination does not by itself guarantee reliable simultaneous fraction recovery.

We study an identifiability-aware framework for designing joint qPCR panels and estimating all cluster fractions simultaneously. Using discovery cells only, we construct reference-normalized, cluster-

specific mean-expression signatures and select a bounded number of non-reference marker genes for each cluster. The forward-selection objective combines target-versus-background discrimination with a penalty for redundancy among selected genes and a reward for identifiable joint signatures. Given a selected signature matrix, held-out pseudobulk mixtures are quantified with non-negative least squares, followed by normalization of the coefficients to enforce compositional closure. This design explicitly connects assay construction to the geometry of the joint deconvolution problem rather than treating marker selection and fraction estimation as unrelated steps.

We verify the framework using discovery and holdout cells from a PBMC single-cell dataset. Independently generated pseudobulk mixtures vary the target abundance and the composition of the background populations, including rare-target and high-abundance regimes. Evaluation includes discrimination and quantitative fraction-recovery metrics, abundance-stratified error, compositional closure diagnostics, and cross-cluster confusion. Bootstrap intervals were not successfully established as a valid completed comparison and are therefore left for future work. We also retain one-vs-rest panel and best-single-marker procedures as reference configurations and perform ablations that remove the redundancy and identifiability terms from the selection objective. This evaluation is designed to distinguish recovery of fractions from mere target detection and to expose failure modes caused by poorly identified signatures.

Our contributions are:

- We formulate compact qPCR panel design and simultaneous cell-fraction estimation as a coupled compositional deconvolution problem.
- We construct discovery-only, reference-normalized joint signatures while excluding reference genes from marker selection.
- We introduce a per-cluster forward-selection procedure that combines discrimination, redundancy control, and joint-signature identifiability.
- We use non-negative least squares with simplex normalization to obtain non-negative fraction estimates whose total is constrained to one.
- We provide an evaluation protocol spanning varied abundance and background-composition regimes, with quantitative, calibration, closure, and cross-cluster confusion diagnostics.

The resulting framework provides a basis for designing small, interpretable assays when the available cell types are not equally distinguishable. Its scope also makes clear several directions for future work, including independently assembled baseline predictions, simulations that more strongly perturb related-cluster similarity, and experiments that report panel-selection and deconvolution outputs without shared metric containers. Such extensions are needed to determine when structural identifiability criteria produce measurable gains beyond discrimination alone and to characterize empirical limits of quantification for closely related or rare populations.

2 RELATED WORK

Bulk cell-fraction deconvolution methods typically treat the cell-type expression signatures as a fixed reference and estimate proportions by solving a regression or constrained linear inverse problem. That formulation is closely related to ours through the linear mixture model and the use of non-negative coefficients, but differs in the measurement regime: our method must first select a compact, non-reference qPCR panel from discovery cells and then apply the resulting signature matrix to independently generated holdout mixtures. Consequently, methods designed for broad bulk-expression profiles are informative conceptual precedents but are not directly applicable experimental baselines here unless they can operate on the same discovery-derived panel and return the complete fraction vector under the same compositional constraints.

Marker-selection and qPCR-panel procedures commonly optimize target-versus-background discrimination or prediction for one population at a time. The one-vs-rest and best-single-marker procedures retained in our experiment represent this alternative: they select genes and fit a target-specific predictor, but do not construct one jointly identifiable signature for all clusters. Such procedures are therefore applicable reference configurations for target detection and marginal fraction prediction, while their objective does not directly penalize redundancy across clusters or assess whether all cell types can be separated simultaneously.

Constrained mixture estimators impose biological feasibility through non-negativity and, in some formulations, a sum-to-one constraint. Our NNLS estimator followed by simplex normalization shares these constraints, but the contribution of our method is to couple the estimator to panel construction: the forward-selection objective evaluates discrimination together with cross-panel redundancy and the geometry of the joint signature. The smallest singular value, condition number, and pairwise cosine similarity are used as diagnostics of whether the selected signatures support a stable inverse problem. This distinguishes the method from approaches that constrain the final regression but select markers without considering joint identifiability.

Within the present Problem Setting, the directly applicable empirical comparisons are the implemented one-vs-rest and best-single-marker procedures, because they use the same discovery–holdout split, pseudobulk mixtures, expression-scale reference normalization, and held-out evaluation protocol. Broad deconvolution methods, assays requiring measurements unavailable in the experiment, and methods relying on externally supplied signatures are relevant context but cannot be compared fairly without changing the data or supervision available to the model. The automated-discovery context of this study is orthogonal to these methodological comparisons (Lu et al., 2024).

3 BACKGROUND

3.1 CELL-FRACTION DECONVOLUTION

A mixed biological sample can be represented as a combination of contributions from multiple cell types. Let K denote the number of cell types and let $\pi \in \mathbb{R}^K$ denote their fractions, where $\pi_k \geq 0$ and $\sum_{k=1}^K \pi_k = 1$. For a panel of G genes, let $\mathbf{y} \in \mathbb{R}^G$ be the measured, reference-normalized expression profile of a mixture and let $\mathbf{S} \in \mathbb{R}^{G \times K}$ contain the corresponding cell-type signatures. Under a linear mixing approximation, the observation is modeled as

$$\mathbf{y} \approx \mathbf{S}\pi + \varepsilon, \quad (1)$$

where ε represents sampling, measurement, and model error. Estimating π is a deconvolution problem: the observed mixture is available, whereas the contributions of its component cell types must be inferred.

The fraction vector has two structural constraints. Non-negativity excludes biologically impossible negative cell abundances, while the sum-to-one constraint expresses compositional closure. In the implemented estimator, these constraints are enforced sequentially: first, non-negative least squares solves

$$\tilde{\pi} = \arg \min_{\pi \geq 0} \|\mathbf{y} - \mathbf{S}\pi\|_2^2, \quad (2)$$

and, when the resulting coefficients have positive total mass, they are normalized afterward:

$$\hat{\pi} = \frac{\tilde{\pi}}{\mathbf{1}^\top \tilde{\pi}}. \quad (3)$$

This post-normalization NNLS procedure produces non-negative estimates on the simplex, but it is not generally equivalent to solving least squares with the simplex constraint imposed directly during optimization. Joint estimation is preferable to fitting each cell type independently when the output must represent one coherent composition, because independent regressions can produce estimates that are mutually inconsistent or outside the feasible fraction domain.

3.2 REFERENCE NORMALIZATION AND MARKER SIGNATURES

qPCR assays commonly use reference genes to reduce sensitivity to sample-wide technical variation. In the present expression-scale formulation, a reference-normalized value for gene g in sample i is represented by a difference between its expression and the mean expression of a reference set $\mathcal{R}$:

$$z_{ig} = x_{ig} - \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} x_{ir}. \quad (4)$$

Here, x_{ig} is the measured expression value and $\mathcal{R}$ is selected for relative stability across discovery clusters. This is a normalization proxy on expression values rather than a physical Ct-scale model.

Reference genes are used for normalization only and are excluded from the candidate marker pool so that the assay’s discriminatory capacity is not conflated with its normalization control.

For discovery cells labeled by cell type k , a reference-normalized signature for candidate gene g is the cluster mean

$$S_{gk} = \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} z_{ig}, \quad (5)$$

where $\mathcal{D}_k$ is the discovery subset belonging to type k . The columns of $\mathbf{S}$ describe how the candidate genes distinguish the cell types. If two columns are nearly collinear, distinct fraction vectors can produce similar mixture profiles, making the inverse problem poorly identifiable. Measures such as pairwise cosine similarity, the smallest singular value of a column-normalized signature matrix, and its condition number quantify complementary aspects of this geometry: low similarity and a larger smallest singular value generally indicate more separated signatures, whereas a large condition number indicates sensitivity to perturbations in the measurements.

3.3 DISCOVERY, HOLDOUT, AND PSEUDOBULK EVALUATION

Single-cell data provide observations from individual cells, but a qPCR assay measures a bulk sample. Pseudobulk simulation bridges these scales by sampling cells from cell-type-specific pools, aggregating their expression, and recording the known mixture fractions. A discovery–holdout design uses discovery cells to select reference genes and marker signatures, while independently sampled holdout cells are reserved for generating evaluation mixtures. This separation permits quantitative assessment of generalization without using the evaluation mixtures to construct the signatures. Varying the target abundance and the background composition is particularly important because rare targets, dominant populations, and mixtures of related cell types stress different parts of the deconvolution problem.

3.4 PROBLEM SETTING

We assume that the sample contains K known cell types, that a discovery set provides labels for estimating their reference-normalized signatures, and that a small non-reference gene panel must be selected before evaluating independently generated holdout mixtures. The primary unknown for each holdout sample is the full fraction vector $\boldsymbol{\pi}$, not a separate binary label for one selected target. The method consequently requires a signature matrix with sufficient rank and separation to support joint inversion, while respecting non-negativity and compositional closure. These assumptions distinguish the setting from one-vs-rest marker classification: target discrimination remains useful for selecting informative genes, but it is only one component of a valid simultaneous fraction-estimation procedure.

4 METHOD

Our method separates panel construction from evaluation. Discovery cells are used to rank candidate markers, select reference genes, construct cell-type signatures, and assemble a joint panel. Independently sampled holdout cells are then used to generate pseudobulk mixtures with known fraction vectors. The selected signatures are applied to these mixtures without refitting the panel, and all cell-type fractions are estimated jointly by non-negative least squares (NNLS) followed by normalization to the probability simplex.

We first stratify cells by their Leiden cluster labels and randomly split each cluster into discovery and holdout subsets, retaining 70% of each cluster for discovery subject to the minimum-size constraints implemented in the split procedure. On the discovery cells, we rank genes within each cluster using Scanpy’s Wilcoxon rank-sum procedure. Up to 30 ranked genes per cluster are retained as marker candidates. All subsequent marker and reference selection steps use discovery data only.

We select reference genes from a stability-ranked pool rather than from the marker candidates. For each gene, we compute its overall mean expression, its expression prevalence, and the standard deviation of its cluster-specific discovery means. Genes are ranked by the ratio of between-cluster standard deviation to absolute overall mean, genes expressed in fewer than 20% of cells are excluded, and candidate marker genes are excluded from the reference pool. The first three genes in the resulting

stable pool are used as references. For any expression vector $\mathbf{x}$, normalization subtracts the mean expression of these reference genes:

$$\tilde{\mathbf{x}} = \mathbf{x} - \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} x_r \mathbf{1}. \quad (6)$$

Because the input matrix contains expression values rather than physical qPCR cycle-threshold measurements, this subtraction is used as an expression-scale normalization proxy.

For each cluster k and selected marker gene g , we compute the mean normalized expression over discovery cells assigned to k . Stacking these values gives the signature matrix $\mathbf{S} \in \mathbb{R}^{G \times K}$ introduced in Section 3.4, with one row per unique selected marker and one column per cluster:

$$S_{gk} = \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} \tilde{x}_{ig}. \quad (7)$$

Reference genes are not included as rows of $\mathbf{S}$. The signature is therefore derived entirely from discovery cells and is used as the fixed design matrix for holdout deconvolution.

We select joint panels with a forward procedure that fills each cluster panel independently, up to four markers per cluster. At every step, each unused candidate for the current cluster is evaluated by constructing the complete provisional signature matrix. Candidate scores combine target-versus-background discrimination with penalties or rewards based on redundancy and joint-signature identifiability. The implementation measures identifiability using the normalized signature geometry, including singular-value and condition-number diagnostics, and excludes reference genes from every candidate panel. The full configuration uses weights of 0.20 for both the redundancy and identifiability terms; the precise score components are treated as implementation details rather than as a separately constrained statistical estimator.

To evaluate simultaneous estimation, we generate discovery-derived and holdout-derived pseudobulk mixtures for the joint evaluation pipeline. For each mixture, a target cluster is selected uniformly, and its initial target abundance is selected from 0.5%, 3%, 8%, 25%, or 65% with probabilities 0.15, 0.20, 0.20, 0.25, and 0.20. The remaining abundance is distributed across the other clusters using a Dirichlet draw with concentration parameter 1.0. A second Dirichlet draw produces the realized mixture proportions, which are converted into counts for 100 sampled cells. Cells are sampled with replacement within each cluster and averaged to form the pseudobulk expression vector. The procedure generates 400 discovery-derived mixtures and 300 holdout-derived mixtures. The NNLS estimator is applied to the holdout mixtures using the discovery-derived signature matrix; it does not fit regression parameters on the discovery-derived mixtures.

For a test mixture with normalized marker-expression vector $\tilde{\mathbf{y}}$, we solve

$$\tilde{\boldsymbol{\pi}} = \arg \min_{\boldsymbol{\pi} \geq 0} \|\mathbf{S}\boldsymbol{\pi} - \tilde{\mathbf{y}}\|_2^2. \quad (8)$$

The implementation uses SciPy’s NNLS solver. If the resulting coefficients have positive total mass, they are divided by their sum:

$$\hat{\boldsymbol{\pi}} = \frac{\tilde{\boldsymbol{\pi}}}{\mathbf{1}^\top \tilde{\boldsymbol{\pi}}}. \quad (9)$$

If the coefficient sum is numerically zero, the implementation substitutes the uniform fraction vector. This post-processing enforces non-negativity and compositional closure for every joint prediction.

For comparison, the experiment also constructs one-target-at-a-time panels. For each target cluster, targeted pseudobulks are generated with target fractions sampled across low, medium, and high abundance regimes, and candidate panels are selected by greedy minimization of cross-validated fraction-prediction MAE. A linear regression model maps the reference-normalized panel score to the target fraction, and predictions are clipped to $[0, 1]$. The best individual candidate gene under the same cross-validated criterion is retained as a single-marker comparator. These procedures provide target-specific reference outputs, whereas the joint model uses one signature matrix and one compositional fraction vector for each mixture.

We evaluate each cluster using ROC AUC for thresholded detection, Spearman and Pearson correlation, MAE, RMSE, R^2 , low- and high-abundance MAE, calibration slope and intercept, and

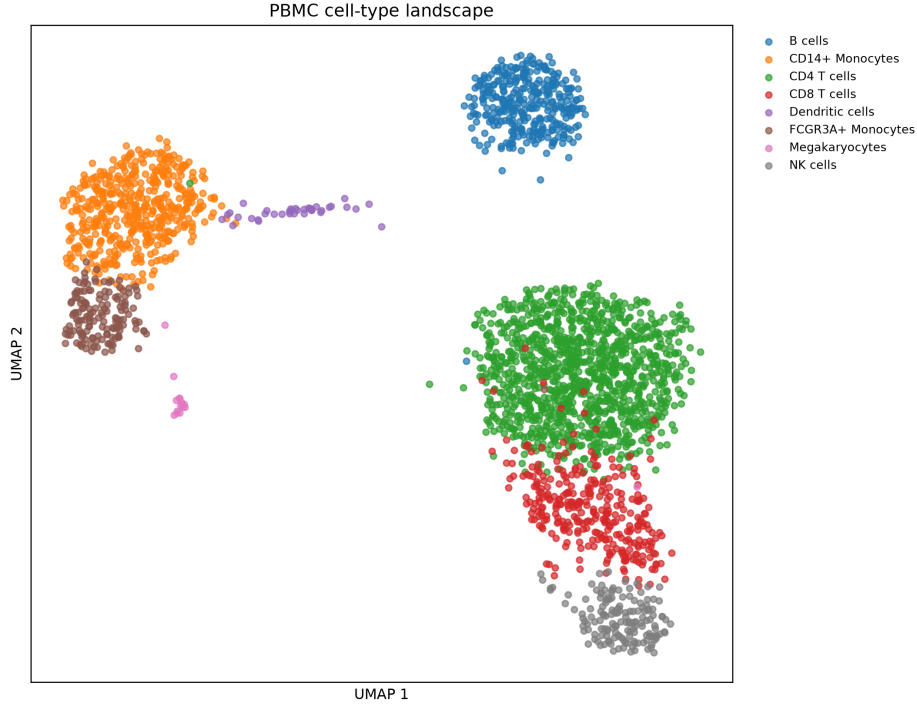


Figure 1: UMAP embedding of the PBMC cells colored by their Leiden cluster labels. The clusters shown here define the discovery–holdout strata and the cell-type pools used for pseudobulk generation.

fraction bias. For joint predictions, we additionally compute the mean absolute and root-mean-square closure error, the rate of approximately non-negative estimates, and the rate of approximately closed fraction vectors. Cross-cluster attribution is summarized by averaging predicted fractions among mixtures grouped by their true dominant cluster. Valid bootstrap intervals for these metrics were not established as a completed comparison and are not reported as experimental results.

5 EXPERIMENTAL SETUP

5.1 DATASET AND CELL-TYPE ORGANIZATION

We evaluate the method on a PBMC single-cell expression dataset supplied as an H5AD file. The completed experiment artifacts identify the file format and the use of Leiden labels, but do not record a public dataset accession or source name; accordingly, no accession is asserted here. The Leiden labels define the cell-type or cluster identities, the discovery–holdout strata, and the cell pools used for pseudobulk generation. The expression matrix is used on its stored expression scale; the implementation does not apply an additional dataset-level transformation before the discovery split, marker ranking, or reference normalization. The resulting cluster organization is shown in Figure 1; this visualization provides dataset context rather than an evaluation of deconvolution accuracy.

The cells are randomly split within each Leiden cluster using a fixed random seed of 42. Approximately 70% of the cells in each cluster are assigned to discovery and the remainder to holdout, with the implementation requiring at least two cells in each partition. Marker ranking, reference-gene selection, and joint signature construction use only the discovery cells. Holdout cells are used only to generate independent test pseudobulks, preventing the evaluation mixtures from contributing to panel construction.

Within each discovery cluster, Scanpy’s Wilcoxon rank-sum procedure ranks genes, and up to 30 genes per cluster are retained as marker candidates. Reference genes are selected from a stability-ranked pool of at most 200 genes, excluding all candidate markers and genes expressed in fewer than 20% of discovery cells; the first three genes in the stable pool are used for normalization.

Reference-normalized expression is computed by subtracting the mean reference-gene expression. The joint forward-selection procedure allocates up to four markers independently to each cluster while evaluating the complete provisional signature. Its redundancy and identifiability weights are both 0.20 in the full configuration.

For the joint model, we generate 400 discovery-derived and 300 holdout-derived pseudobulks, each containing 100 sampled cells. Every mixture selects a target cluster uniformly. The target-fraction design uses point masses at 0.005, 0.03, 0.08, 0.25, and 0.65 with probabilities 0.15, 0.20, 0.20, 0.25, and 0.20, respectively. Background proportions are drawn with a Dirichlet concentration parameter of 1.0, and a second Dirichlet draw produces the realized mixture composition before multinomial sampling. Cells are sampled with replacement within each cluster and averaged to produce each pseudobulk expression profile. The discovery-derived mixtures are generated as part of the evaluation pipeline but are not used to fit regression parameters: NNLS is applied directly to the holdout mixtures using the discovery-derived signature matrix.

The experiment retains two one-target-at-a-time reference procedures. For each target cluster, 400 discovery-derived training pseudobulks and 300 holdout-derived test pseudobulks are generated, each with 100 sampled cells. The target fraction is sampled by first choosing low, medium, or high regimes with probabilities 0.30, 0.45, and 0.25, respectively, and then drawing uniformly from $[0.01, 0.10]$, $[0.10, 0.50]$, or $[0.50, 0.90]$ for the selected regime. Remaining cells are distributed across the other clusters with a Dirichlet draw. Panels are selected greedily by cross-validated fraction-prediction MAE, with a maximum of five markers and a minimum required MAE improvement of 0.002 for each addition. A linear regression maps the normalized panel score to the target fraction, and predictions are clipped to $[0, 1]$. The best single candidate gene is evaluated as a separate comparator. For the joint-selection ablations, we compare discrimination-only selection, discrimination plus redundancy control, and the full objective with both redundancy and identifiability terms active, while keeping the data split, mixture simulation, NNLS estimator, and simplex normalization fixed.

For each cluster, ROC AUC is computed from the continuous predicted fraction or score and does not depend on the detection threshold. The threshold of 0.35 is used to define binary detection labels for threshold-based quantities. Sensitivity at a nominal specificity of 0.95 is calculated separately by selecting a score threshold from the negative examples at that specificity and measuring the fraction of positive examples above it. Fraction recovery is summarized by Spearman correlation, Pearson correlation, MAE, RMSE, and R^2 . We additionally report MAE for fractions at or below 0.10 and at or above 0.50, calibration slope and intercept, and mean fraction bias. For joint predictions, we measure mean absolute and root-mean-square closure error, the proportion of estimates satisfying the non-negativity tolerance of 10^{-8} , and the proportion satisfying the closure tolerance of 10^{-6} . Cross-cluster attribution is summarized by the mean predicted fraction conditioned on the true dominant cluster.

The implementation contains a 200-resample bootstrap routine for per-cluster MAE, RMSE, and Spearman correlation, but valid bootstrap intervals were not established as a completed comparison and are not reported as experimental results. Aggregate metrics are macro-averages across clusters, and panel size is summarized by the mean number of selected genes per cluster. The experimental comparisons are intended to assess joint fraction recovery, detection behavior, abundance-dependent error, signature geometry, compositional closure, and the effect of the implemented selection ablations without using later experimental results to define the setup.

6 RESULTS

We first characterize the selected markers and then evaluate held-out detection and fraction recovery. We compare the existing one-target baseline with the corrected joint configuration, for which each cluster receives its intended marker allocation, while noting that the procedures use different target-abundance sampling schemes: the baseline samples continuously from low, medium, and high intervals, whereas the joint procedure uses specified point masses. We then examine abundance-dependent error, cross-cluster attribution, compositional closure, and the discrimination-only and redundancy-aware selection ablations.

The selected-marker expression plots provide biological context for the two principal panel constructions. The baseline marker panel is shown in Figure 2, while the corrected joint panel is shown in

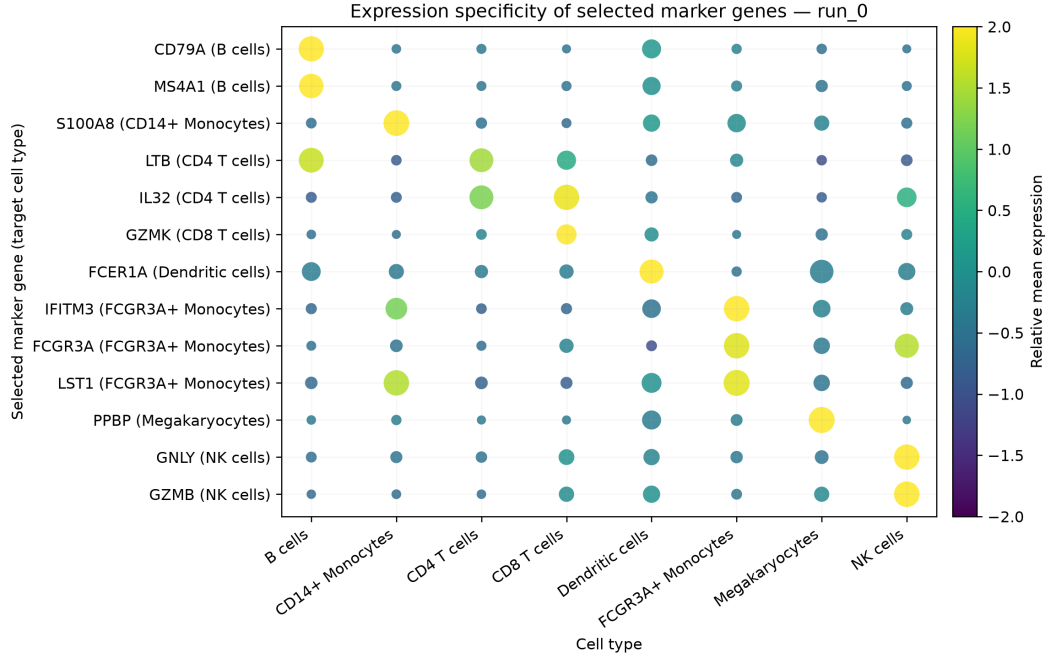


Figure 2: Expression specificity and prevalence of the selected baseline marker genes across Leiden clusters. Rows identify genes and their target clusters; dot size represents expression prevalence and color represents relative cluster mean expression.

Figure 3. These plots display relative cluster mean expression and expression prevalence for each selected gene, allowing cluster specificity to be assessed directly. The joint panel contains an average of 3.75 selected genes per cluster, compared with 1.625 genes for the baseline panel.

On the baseline’s independently generated holdout pseudobulks, the one-target panel achieved macro ROC AUC of 0.9873, macro Spearman correlation of 0.9539, macro MAE of 0.0588, and macro RMSE of 0.0742; its low- and high-abundance MAEs were 0.0432 and 0.0791, respectively. On the joint procedure’s holdout pseudobulks, the corrected configuration achieved macro ROC AUC of 0.9814, macro Spearman correlation of 0.8527, macro Pearson correlation of 0.9012, macro MAE of 0.0455, macro RMSE of 0.0669, and macro R^2 of 0.7694. Its low- and high-abundance MAEs were 0.0340 and 0.1438, respectively. These values describe the completed outputs but do not constitute an unqualified head-to-head comparison, because the baseline and joint procedures use different target-abundance distributions and the joint output fields do not provide an independent baseline prediction.

The held-out ROC curves in Figure 4 provide the per-cluster detection view corresponding to the reported macro ROC AUC. The predicted-versus-true fraction plots in Figure 5 show recovery separately for each cluster and abundance regime, while the residual plots in Figure 6 show how signed errors vary with true abundance. Together, these figures support the aggregate finding that error is not uniform across the abundance range: low-fraction recovery is comparatively accurate, whereas high-abundance samples exhibit larger deviations and broader residual behavior.

An initial joint configuration produced macro MAE 0.1778, macro RMSE 0.2428, and macro $R^2 = -2.3905$, with macro ROC AUC 0.5733 and high-abundance MAE 0.4920. Its average panel size was 0.5 genes per cluster, so the joint signature did not contain enough rows to identify all cluster fractions reliably. After correcting the marker allocation so that panels were filled independently for each cluster, macro MAE decreased to 0.0455, macro RMSE to 0.0669, and macro R^2 increased to 0.7694. This comparison demonstrates the practical importance of allocating sufficient markers to every cluster; it should not be interpreted as an isolated test of the structural selection terms.

The joint fraction and closure diagnostic in Figure 9 assesses both quantitative recovery and the sum-to-one constraint, while Figure 10 summarizes predicted fractions conditional on the true dominant

cluster. NNLS followed by coefficient normalization yields predictions designed to be non-negative and closed by construction. The discovery-derived signature and cosine-similarity diagnostic in Figure 7 connects the selected markers to the identifiability hypothesis, and the cross-configuration structural comparison in Figure 8 summarizes the recorded error and signature-geometry quantities across the corrected joint configurations. These diagnostics are important because aggregate per-cluster error alone does not reveal whether errors are concentrated among particular cell types or whether the selected signatures are sufficiently separated for joint inversion.

The absolute-error-versus-abundance plot in Figure 11 gives a more detailed view of the empirical quantification range. It is consistent with the aggregate stratification: rare and low-abundance fractions can be recovered with smaller absolute errors than dominant fractions in the completed evaluation. The results therefore support a practical limitation of the current setup for high-abundance compositional estimation, where allocating the remaining fraction among background populations may be harder than detecting the dominant target.

The discrimination-only and redundancy-aware ablations produced the same reported aggregate results as the corrected full configuration to the displayed numerical precision: macro ROC AUC 0.9814, macro Spearman correlation 0.8527, macro Pearson correlation 0.9012, macro MAE 0.0455, macro RMSE 0.0669, macro R^2 0.7694, low-abundance MAE 0.0340, and high-abundance MAE 0.1438. Their mean panel size was also 3.75 genes per cluster. The cross-configuration summary in Figure 12 therefore documents a stable set of reported predictive metrics, but it does not establish that redundancy control or the identifiability reward improves performance.

The baseline outputs report a macro panel-minus-single-marker AUC difference of 0.0035, whereas the corrected joint outputs report a difference of 0.0. However, the joint panel and single-marker prediction fields are identical in the recorded joint outputs, so the paired comparison in Figure 13 cannot be treated as independent evidence that a multi-gene joint panel outperforms a best-single-marker baseline. This limitation also applies to the ablation comparisons because the reported prediction fields are duplicated for the joint configurations, and the differing abundance designs further limit direct numerical comparison between baseline and joint results.

Overall, the completed experiments show that correcting the per-cluster marker allocation was necessary for reliable joint fraction recovery in this setup. The corrected configuration achieved strong aggregate detection and substantially improved fraction-recovery metrics relative to the underidentified preliminary configuration, but high-abundance estimation remained less accurate. The selection ablations were numerically indistinguishable from the full objective, and the stored baseline fields prevent a valid independent joint-versus-single-marker comparison. Consequently, the experiments establish recovery from an allocation failure and characterize the remaining abundance-dependent limitations, but they do not yet demonstrate an ablation-sensitive benefit from redundancy-aware or identifiability-aware selection.

7 CONCLUSIONS AND FUTURE WORK

We presented an identifiability-aware approach for designing compact qPCR panels and estimating cell-type fractions jointly in mixed samples. The method uses discovery-only, reference-normalized signatures, excludes reference genes from marker selection, allocates markers independently to each cluster, and combines target discrimination with redundancy and signature-identifiability criteria. Held-out pseudobulk mixtures are quantified with NNLS followed by simplex normalization, producing non-negative fraction estimates that sum to one.

The completed experiments showed that correct per-cluster panel allocation was essential in this setting. Restoring the intended allocation improved macro MAE from 0.1778 to 0.0455, macro RMSE from 0.2428 to 0.0669, and macro R^2 from -2.3905 to 0.7694; the corrected configuration reached macro ROC AUC of 0.9814. Quantification remained more difficult for high-abundance fractions than for fractions at or below 0.10, with MAE of 0.1438 and 0.0340, respectively. However, the full objective and its discrimination-only and redundancy-aware ablations produced indistinguishable reported aggregate metrics, and the saved joint and single-marker prediction fields were duplicated. Thus, the experiments demonstrate recovery from an underidentified allocation failure, but do not establish a comparative advantage for the structural selection terms or a valid joint-panel advantage over the single-marker baseline.

Future work should first separate baseline and joint prediction containers throughout evaluation so that panel, single-marker, and one-vs-rest comparisons remain independent. It should then verify whether objective terms change the selected panels and the signature geometry, rather than relying only on aggregate prediction metrics. More challenging simulations should vary related-cluster similarity and background composition explicitly, include additional rare-target regimes, and evaluate whether improvements persist across independent discovery–holdout splits. Reporting valid bootstrap intervals for independently assembled methods and testing panels on experimentally measured mixtures would further clarify the uncertainty and practical limits of compositional qPCR deconvolution.

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

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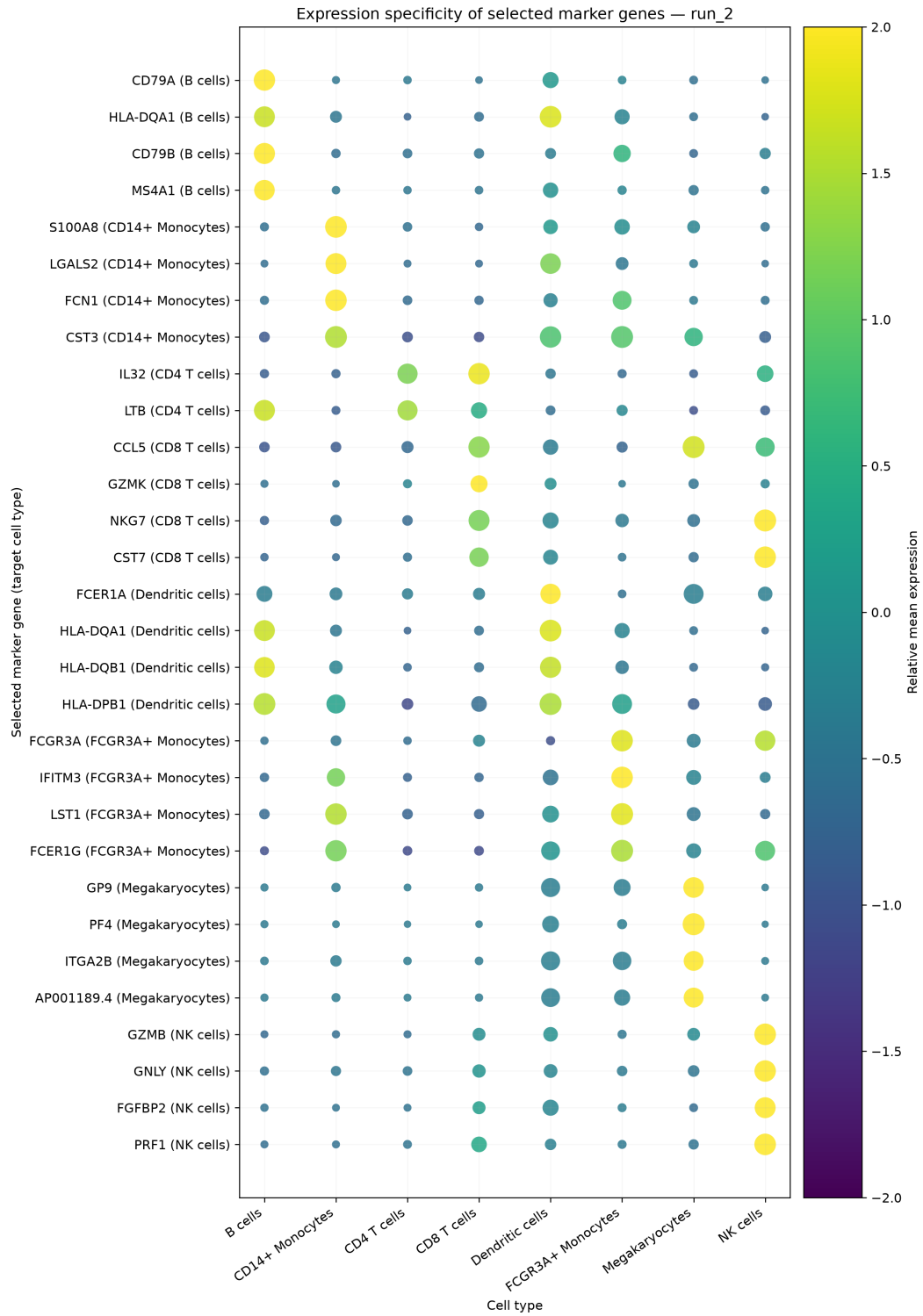


Figure 3: Expression specificity and prevalence of the selected corrected joint-panel markers across Leiden clusters. The panel is constructed from discovery cells and excludes reference genes.

Held-out cell-type detection performance — run_2

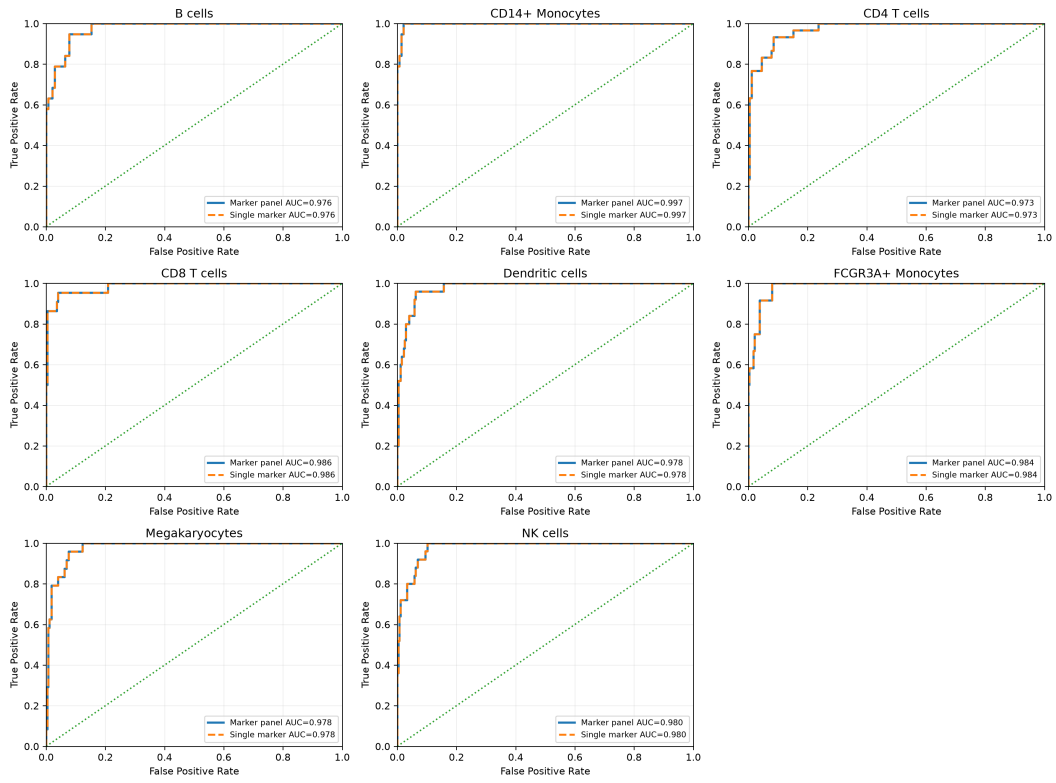


Figure 4: Per-cluster ROC curves for the corrected joint configuration on held-out pseudobulk mixtures. Curves are computed from continuous predicted fraction scores; the displayed AUC values summarize threshold-independent discrimination.

Quantitative cell-fraction prediction — run_2

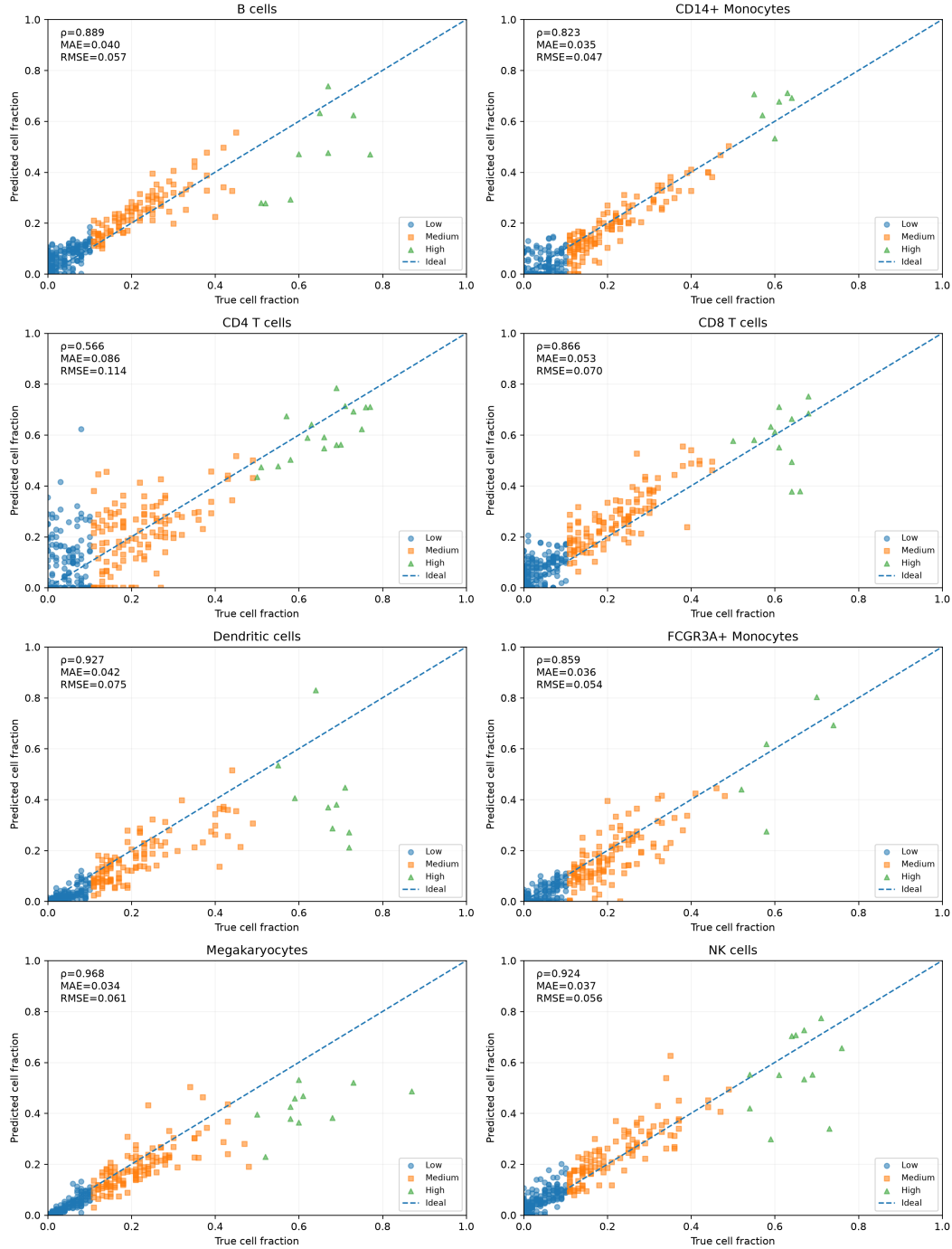


Figure 5: Predicted versus true cell fractions for the corrected joint configuration on held-out pseudobulk mixtures. Point shapes denote low, medium, and high abundance regimes; the dashed line denotes exact recovery.

Prediction residuals across abundance regimes — run_2

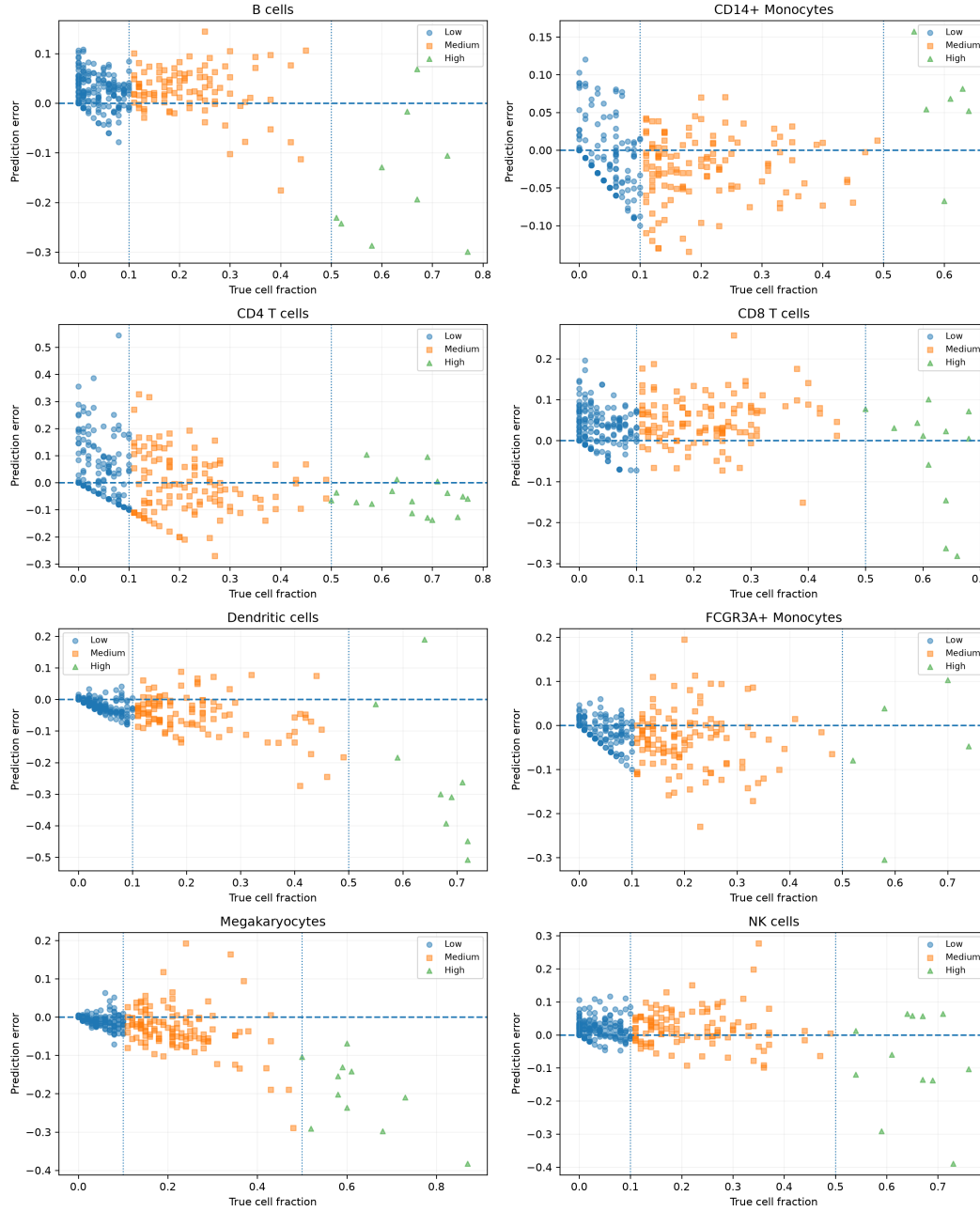


Figure 6: Signed fraction-estimation residuals versus true abundance for the corrected joint configuration. Vertical reference lines indicate the low- and high-abundance thresholds.

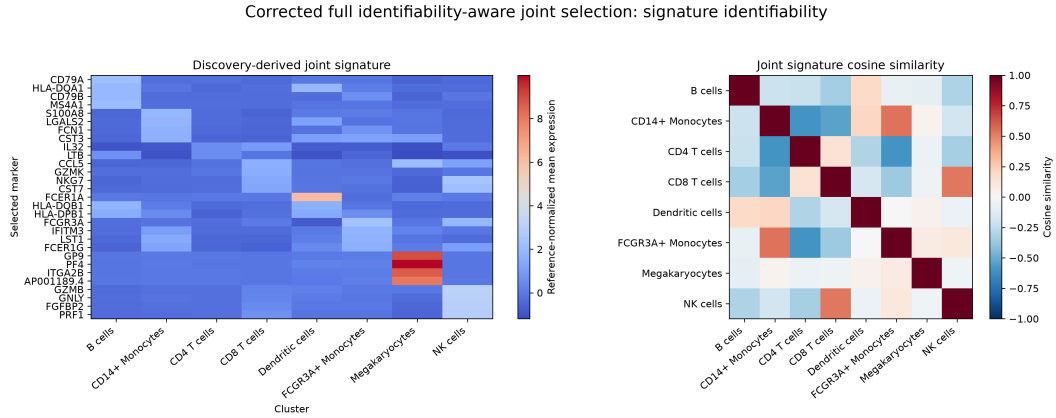


Figure 7: Discovery-derived joint signature and pairwise cosine similarity for the corrected configuration. The two views provide complementary diagnostics of marker-expression structure and separation among cluster signatures.

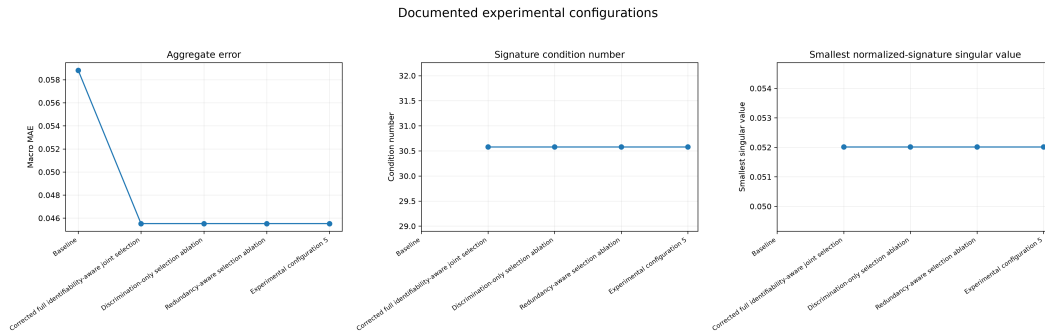


Figure 8: Recorded aggregate error and signature-geometry quantities across the completed corrected joint configurations and selection ablations.

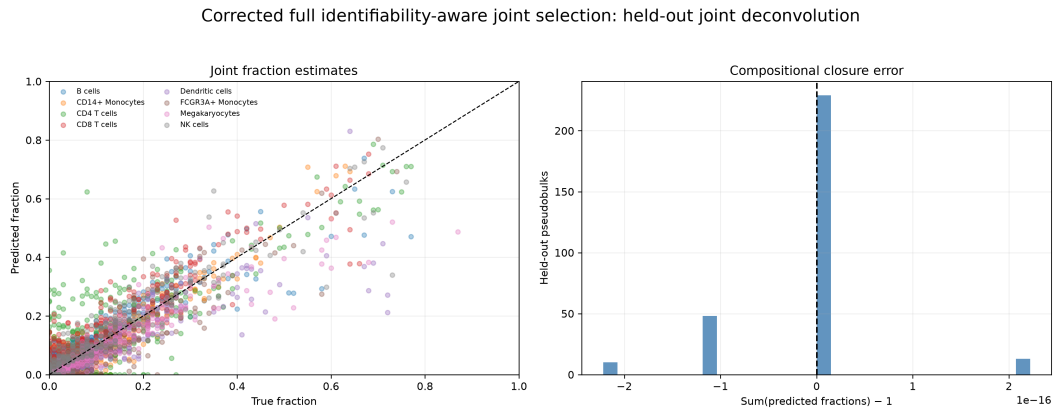


Figure 9: Held-out joint fraction predictions and the distribution of predicted-fraction sum minus one for the corrected configuration. The left panel shows quantitative recovery and the right panel shows compositional closure error.

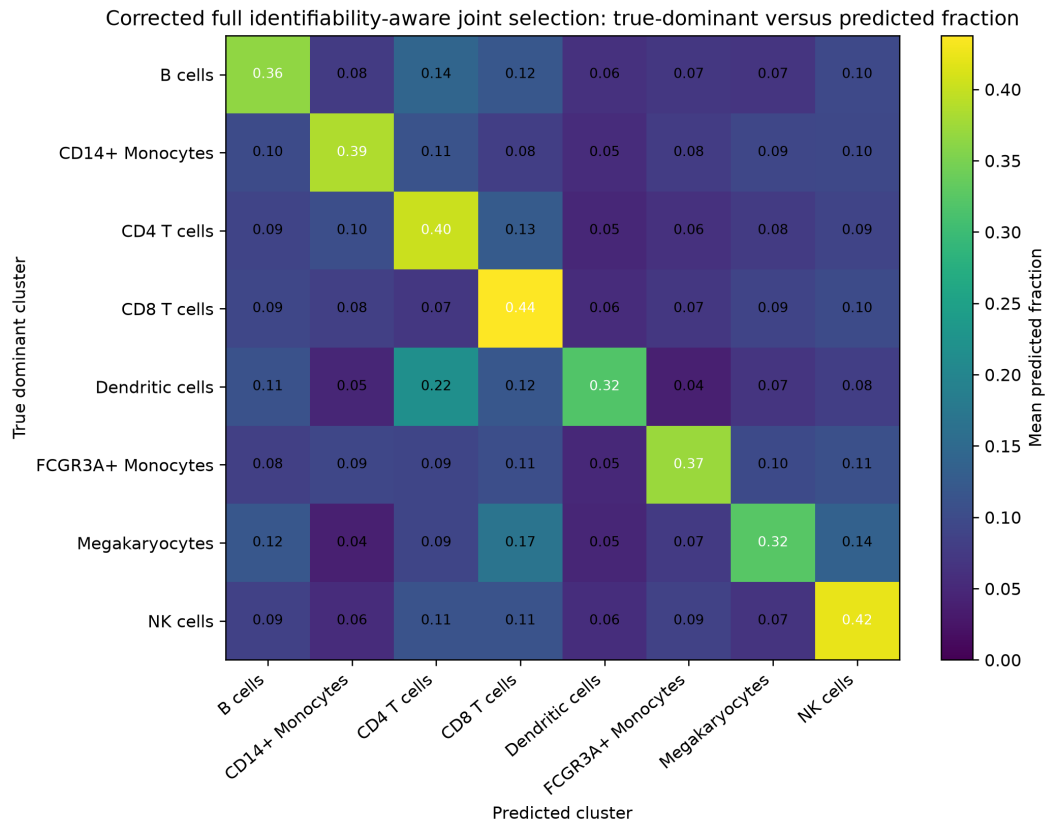


Figure 10: Mean predicted fractions grouped by the true dominant cluster for the corrected joint configuration. Diagonal entries represent allocation to the true dominant cluster; off-diagonal entries represent cross-cluster attribution.

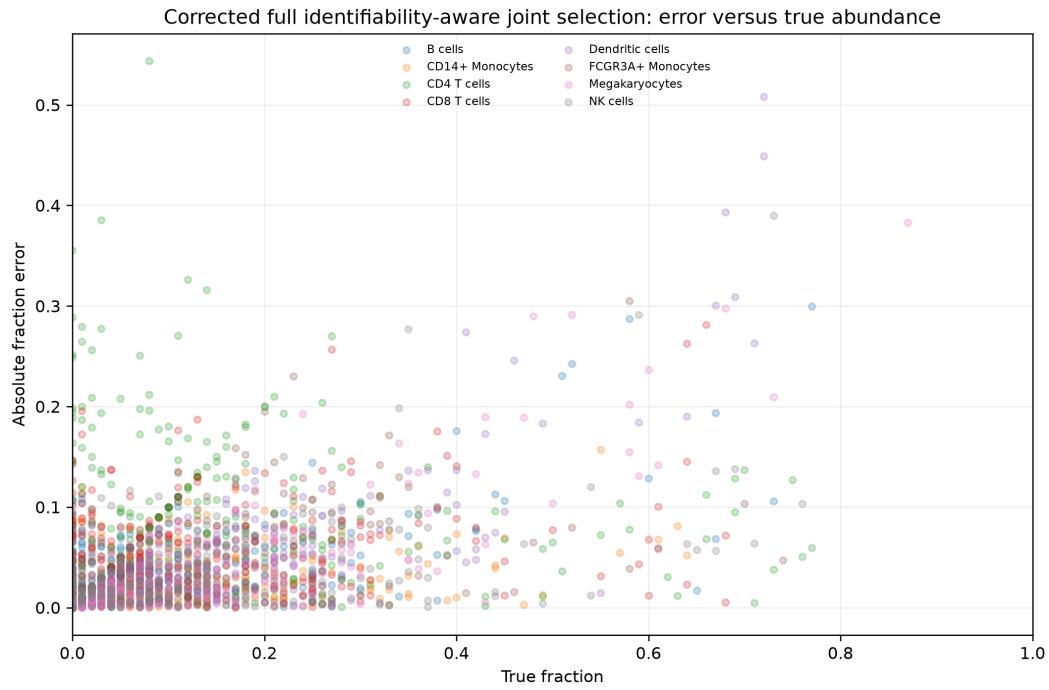


Figure 11: Absolute fraction-estimation error as a function of true abundance for the corrected joint configuration. Each color identifies a cluster, revealing abundance-dependent and cluster-dependent quantification difficulty.

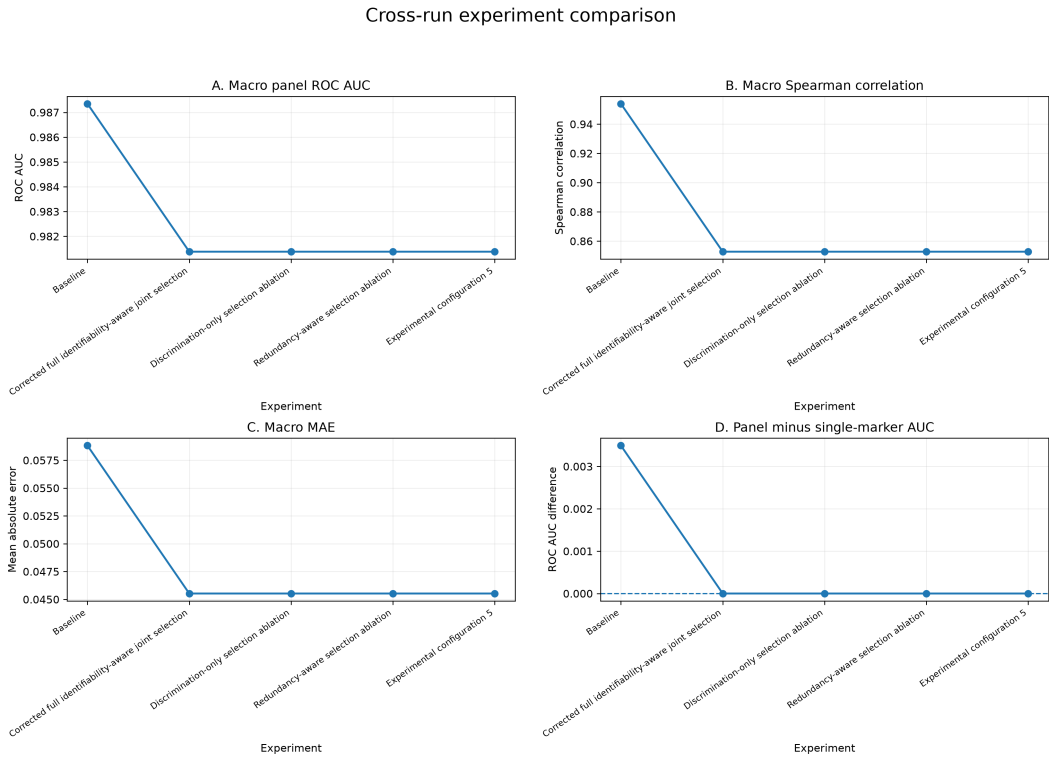


Figure 12: Reported aggregate metrics across the completed experimental configurations, including detection, rank correlation, fraction error, and panel–single-marker AUC difference. The corrected joint and ablated configurations are numerically indistinguishable in these outputs.

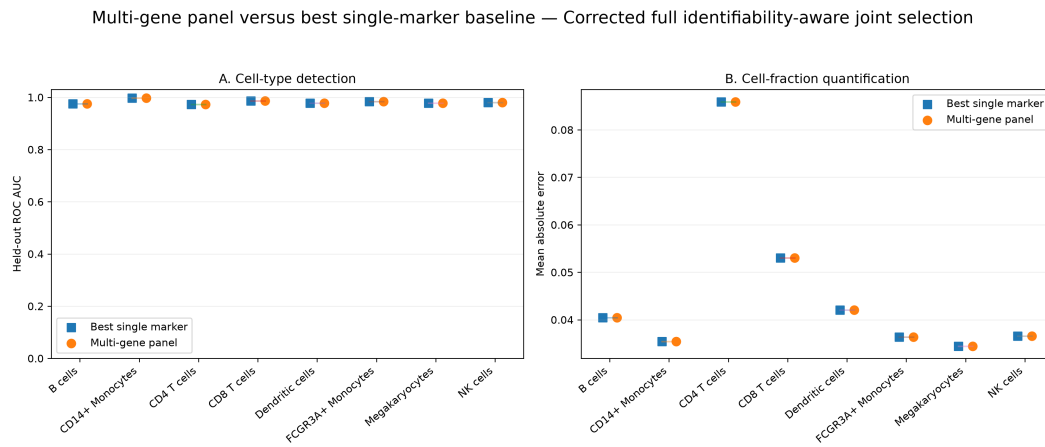


Figure 13: Stored per-cluster comparison between the panel and single-marker prediction fields for the corrected joint configuration. The identical reported fields make this a diagnostic of the recorded outputs rather than an independent baseline comparison.